

INNOVA JUNIOR COLLEGE  
JC 2 PRELIMINARY EXAMINATION  
in preparation for General Certificate of Education Advanced Level  
**Higher 2**

CANDIDATE  
NAME

CLASS

INDEX NUMBER

**BIOLOGY**

**9744/04**

Paper 4 Practical

**17 August 2017**

**2 hours 30 minutes**

Candidates answer on the Question Paper.

**READ THESE INSTRUCTIONS FIRST**

Write your name and class on all the work you hand in.  
Give details of the practical shift and laboratory, where appropriate, in the boxes provided.  
Write in dark blue or black pen on both sides of the paper.  
You may use an HB pencil for any diagrams or graphs.  
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided in the Question Paper.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.  
The number of marks is given in the brackets [ ] at the end of each question or part question.

|                   |
|-------------------|
| <b>Shift</b>      |
|                   |
| <b>Laboratory</b> |
|                   |

| For Examiner's Use |    |
|--------------------|----|
| 1                  | 24 |
| 2                  | 17 |
| 3                  | 14 |
| Total              | 55 |

This document consists of **13** printed pages and **1** blank page.



Answer **all** questions

- 1 The enzyme **E** catalyses the hydrolysis of sucrose to produce fructose and glucose.

The products of the hydrolysis of sucrose will reduce potassium permanganate from purple to colourless as follows:

purple  $\longrightarrow$  colourless

You are required to investigate the effect of sucrose concentration on the progress of this enzyme-catalysed reaction by finding the time taken for the decolourization of potassium permanganate.

You are provided with

- 1% enzyme solution, labelled **E**
- 10% sucrose solution, labelled **S1**
- 2%, 4%, 6%, 8%, 10% glucose solution, labelled **G1 – G5**
- 1 mol dm<sup>-3</sup> sulfuric acid, labelled **A**
- 0.01% potassium permanganate solution, labelled **P**
- distilled water labelled, **W**

 **Sulfuric acid and potassium permanganate are harmful. If any comes into contact with your skin wash immediately under cold water. It is recommended that you wear safety goggles.**

Proceed as follows:

- 1 Prepare an appropriate volume of 5% sucrose and label it **S2**.
- 2 Put 1 cm<sup>3</sup> of **A** into a test-tube.
- 3 Add 1 drop of **P** into the same test-tube. Gently shake to mix.
- 4 Add 1 cm<sup>3</sup> of **G1** to the test-tube. Start the stopwatch.
- 5 Record the time taken for **P** to decolourise in step **12**.
- 6 Repeat step **2** to **5** for **G2** to **G5**. You may perform the test simultaneously.
- 7 Put 5 cm<sup>3</sup> of **S1** into a small beaker.
- 8 Add 1 cm<sup>3</sup> of **E** into the small beaker containing **S1**.
- 9 Stir to mix the solutions. Allow the reaction to take place for 2 minutes.
- 10 Perform step **2** to **5** for reaction mixture of **E** and **S1**.
- 11 Repeat step **7** to **10** for **S2** you have prepared in step **1**.

- 12** Record your data in a suitable format in the space provided below. If **P** does not decolourise, record 'more than 600'.

[5]

- (a)** **(i)** Using your results in step **12**, estimate the concentration of reducing sugar in the reaction mixture with

**S1** ..... [1]

**S2** ..... [1]

- (ii)** With reference to enzyme action, explain your observations for **S1** and **S2**.

.....  
.....  
.....  
..... [2]

- (b)** Describe a suitable control for this investigation.

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.....  
..... [2]

- (c)** Identify **two** significant sources of error in this investigation.

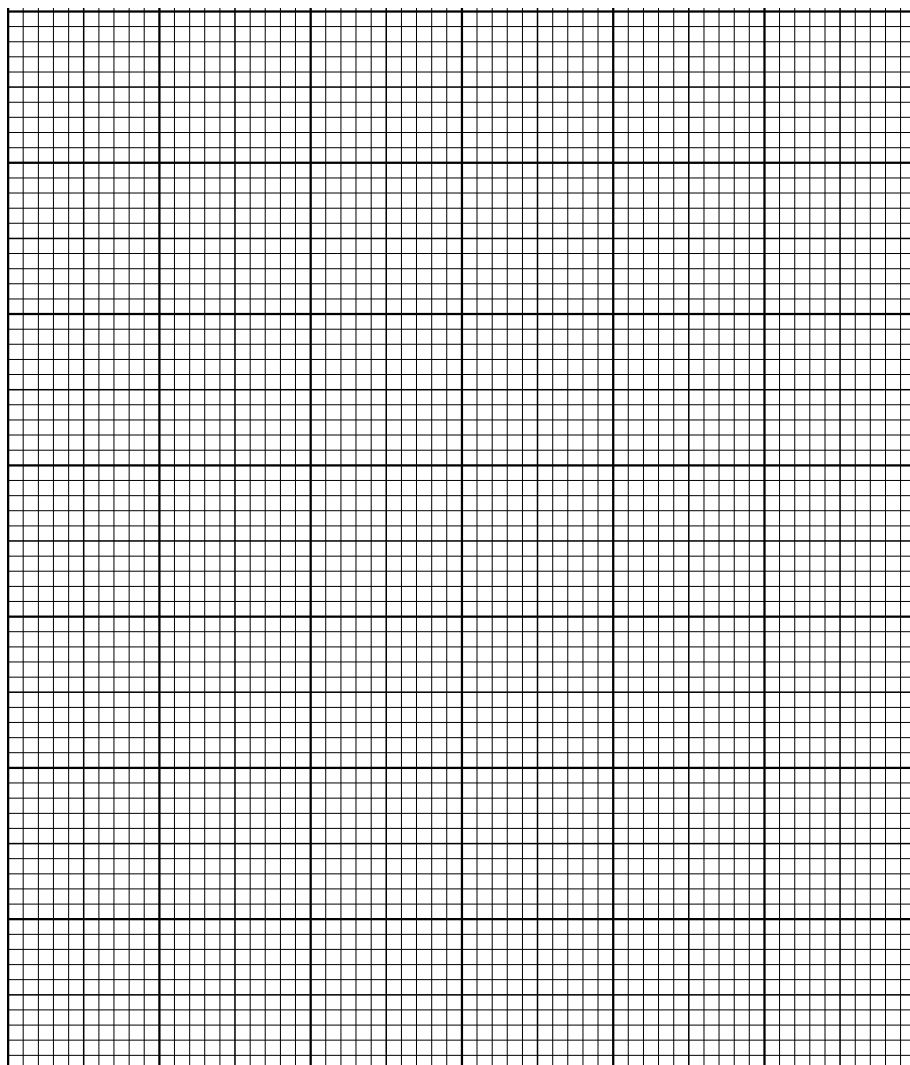
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.....  
..... [2]

- (d) Table 1.1 shows the results for a similar investigation which measured the mass of reducing sugars produced over a period of 400 seconds.

**Table 1.1**

| time / s | mass of reducing sugars / mg ml <sup>-1</sup> |
|----------|-----------------------------------------------|
| 60       | 0.32                                          |
| 120      | 0.64                                          |
| 180      | 0.95                                          |
| 300      | 1.55                                          |
| 400      | 2.05                                          |

- (i) Plot a graph of the data shown in Table 1.1.



[4]

- (ii) Using your graph, find the rate of hydrolysis of the sucrose.  
Show on your graph where you took the readings to calculate the rate. [1]
- Show all working in your calculation.

rate of enzyme activity ..... [1]

- (iii) Describe how results shown in Table 1.1 can be obtained if you are provided with Benedict's solution, 3 mg ml<sup>-1</sup> reducing sugar solution and a colorimeter.

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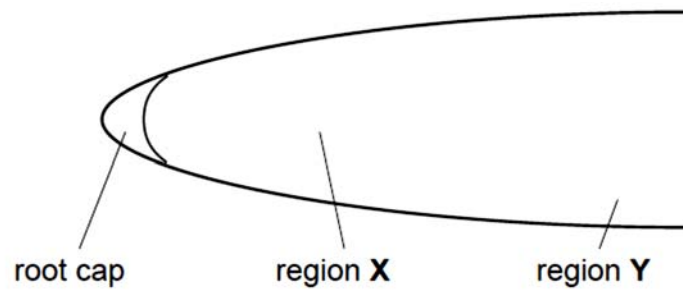
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..... [5]

[Total: 24]

**2** **K1** is a stained, longitudinal section of a young root tip.

Use your microscope to examine carefully the regions labelled **X** and **Y** in Fig. 2.1.



**Fig 2.1**

**(a)** Make a large, labelled, high-power drawing of a single cell at

**(i)** metaphase

Magnification =

[3]

**(ii)** anaphase

Magnification =

[3]

- (b)** **(i)** Using the eyepiece graticule fitted in the eyepiece lens of your microscope, state the objective lens you are using and the number of eye piece divisions equivalent to length of the cell you have drawn in **(a)(i)** and **(ii)**.

objective lens .....

number of eyepiece graticule divisions for **(a)(i)** = ..... [1]

number of eyepiece graticule divisions for **(a)(ii)** = ..... [1]

- (ii)** Using the stage micrometer, determine the length of one division on your eyepiece graticule at the objective stated in **(b)(i)**.

Show the measurements you have made and your working.

[2]

- (iii)** Using your results in **(b)(i)** and **(ii)**, find the actual length, in  $\mu\text{m}$ , of the length of the cells that you have drawn in **(a)(i)** and **(ii)**.

Show your working in the space provided.

[2]

- (iv)** Indicate the actual length of the cells in an appropriate manner on your diagram in **(a)**. [1]

- (v)** Calculate and state the magnification of your drawing in **(a)**. Show your working.[2]

- (c)** Describe **two** differences observed between cells at region **X** and **Y**.

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..... [2]

[Total: 17]

- 3 Respiratory quotient (RQ) is a measurement of the ratio of carbon dioxide given out to oxygen taken in. The RQ value acts as an indication of the respiratory substrate used.

$$\text{RQ} = \frac{\text{CO}_2 \text{ given out}}{\text{O}_2 \text{ taken in}}$$

Carbohydrates often give a RQ of 1.0, while protein and fats give 0.8 and 0.7, respectively.

Yeast are unicellular eukaryotic organisms that respire using a range of substrates.

You are required to plan, but not carry out, an experiment to investigate the RQ when yeast is metabolising different carbohydrates in respiration.

You must use:

- active yeast suspension in small conical flask
- 5% glucose
- 5% sucrose
- rubber bung with delivery tube
- soda lime in syringe
- T-shaped connecting tube
- capillary tube
- blue ink

Fig. 3.1 shows part of the experimental setup.

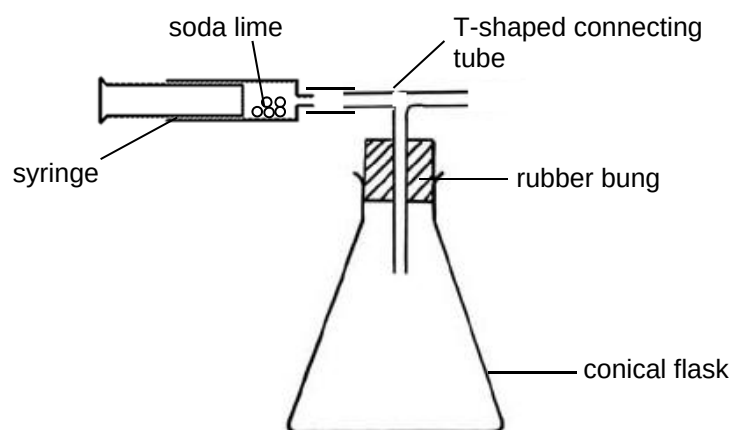


Fig. 3.1



You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware, e.g. test-tubes, boiling tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.
- syringes
- timer, e.g. stopwatch
- thermostatically regulated electrical water bath

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it
- be illustrated by relevant diagram(s), if necessary, to show, for example, the arrangement of the apparatus used
- identify the independent and dependent variables
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and repeatable as possible
- include layout of results tables and graphs with clear headings and labels
- use the correct technical and scientific terms
- include reference to safety measures to minimise any risks associated with the proposed experiment

[Total: 14]

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